

# Majorana Modes in the Machine

Week 6: From One Measurement to a Phase Diagram

Dobromir Stoev  
Edgar Harutyunyan  
Guilherme Schewtschik  
Jaskaran Singh  
Zhengyi Liu  
Zhenming Shi

Advanced Seminar – AI-Augmented Theoretical Physics

August 10, 2026

# Week 6 Objective

## Goal

Turn the Week 5 proof-of-principle into a diagnostic: sweep  $\mu$  and ask whether a gate-measurable non-local correlator reconstructs the topological phase boundary.

- Week 5 measured one sweet-spot example at  $t = \Delta$ ,  $\mu = 0$ .
- Week 6 varies  $\mu$  across trivial, critical, and topological regimes.
- The target observable remains parity-preserving and non-local.
- The emphasis now shifts from solving the model to diagnosing phases operationally.

# The Observable We Keep

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$$

- Single-edge operators flip fermion parity, so their expectation vanishes in a parity eigenstate.
- Unlike Ising-type order parameters, topology cannot generally be detected locally.
- The Jordan–Wigner mapping converts an edge-to-edge Majorana correlator into a non-local Pauli string.
- The two-edge string preserves parity and probes coherent long-range edge correlations.
- For  $L = 4$ :

$$O_{\text{edge}} = X_0 Z_1 Z_2 X_3.$$

# Symmetry Breaking vs Topology

## AFM transverse-field Ising chain

$$H = J \sum_j Z_j Z_{j+1} - h \sum_j X_j, \quad J > 0$$



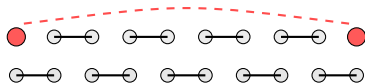
$$m_{\text{stag}} = \frac{1}{L} \sum_j (-1)^j Z_j \neq 0$$

Two locally distinguishable Neel states; local domain-wall excitations.

**Same Ising-like algebra, different physics:** local Neel order versus parity sectors and edge-to-edge coherence.

## Kitaev chain at $t = \Delta$

$$H = t \sum_j Y_j Y_{j+1} - \frac{\mu}{2} \sum_j Z_j$$



$$P = \prod_j Z_j, \quad O_{\text{edge}} = X_0 Z \cdots Z X_{L-1}$$

Low-energy states differ by parity; Majorana edge degeneracy is non-local.

# Measurement Protocol

## Circuit-level measurement

- Current validation: exact even-parity ground state from diagonalization.
- Hardware target: parity-constrained VQE state preparation.
- Apply Hadamards on qubits carrying  $X$  factors.
- Measure all qubits and multiply outcomes to estimate  $\langle O_{\text{edge}} \rangle$ .

## Shot estimator ( $N = 4096$ )

Each shot returns an eigenvalue  $m_s = \pm 1$ .

$$\langle O \rangle \approx \frac{1}{N} \sum_{s=1}^N m_s, \quad \sigma \approx \sqrt{\frac{1 - \langle O \rangle^2}{N}}$$

For any  $\pm 1$  estimator with  $N = 4096$ ,  $\sigma \leq 1/\sqrt{N} \approx 0.016$ . Error bars are omitted from the sweep plot for clarity.

# Expected Physical Behavior

## Topological

- Edge modes separated
- Finite string order
- Coherent boundaries

$$|\mu| < 2t$$

## Critical

- Bulk gap closes
- Correlation length diverges
- Edge delocalization

$$|\mu| = 2t$$

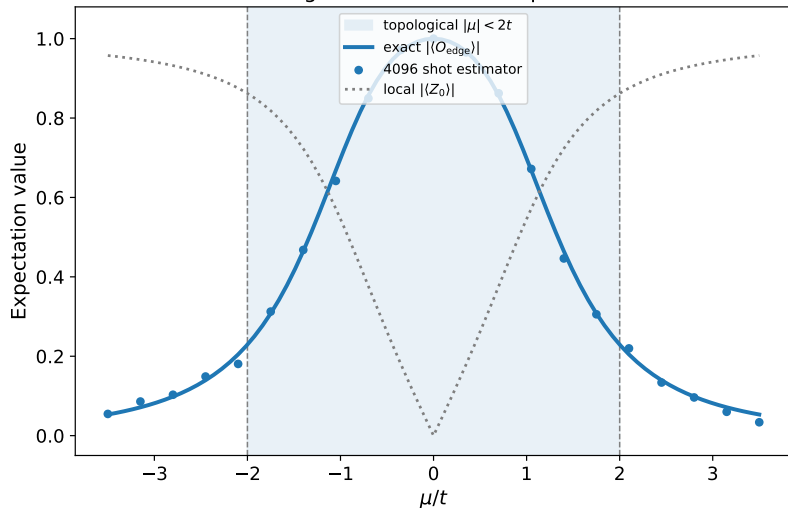
## Trivial

- No protected localized zero modes
- Parity gap becomes finite
- Edge string becomes small

$$|\mu| > 2t$$

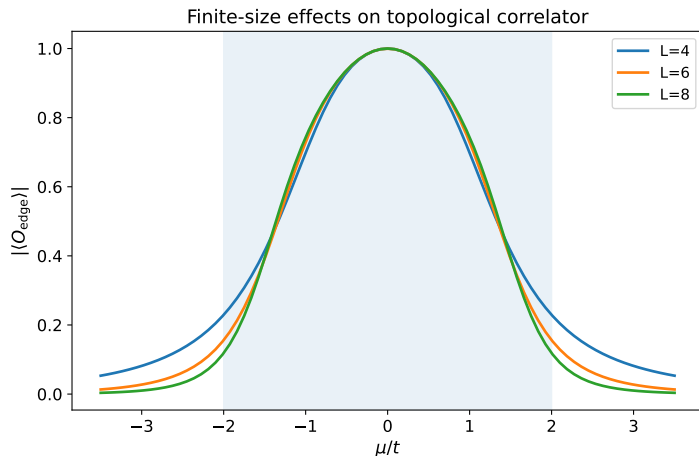
# Numerical Phase Sweep

Non-local diagnostic of the Kitaev phase transition



- $L = 4, t = \Delta = 1$ ;  
sweep  $\mu$
- Edge string:  
measurable diagnostic
- Parity gap: ED  
cross-check

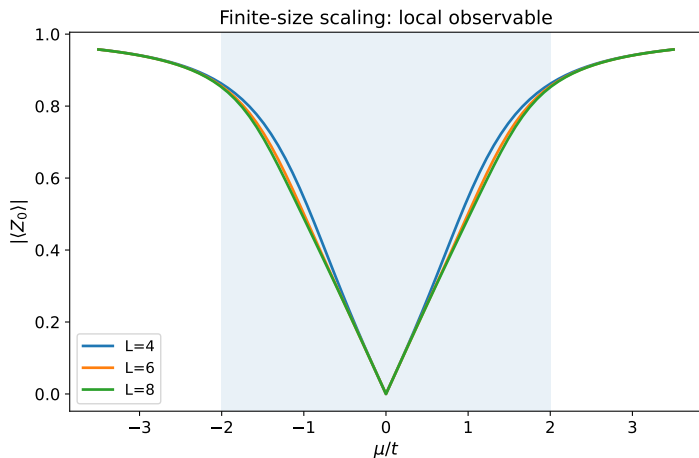
# Finite-Size Scaling: Non-Local Observable



- Small  $L$ : smooth crossover
- Larger  $L$ : sharper boundary
- $L = 4, 6, 8$  remain far from the thermodynamic limit
- The sharpening *trend*, not convergence, is the result



# Finite-Size Scaling: Local Observable



- Local observable shows weak dependence on system size
- No sharpening near the phase transition
- Confirms that topology is not locally detectable

# Implementation Step

```
for mu in mu_values:
    H = kitaev_qubit_hamiltonian(L, t, mu, delta)
    psi = even_parity_ground_state(H)    # exact diagonalization, NOT VQE
    value = <psi| X0 Z1 ... Z(L-2) X(L-1) |psi>
    shot_value = sample_pm_one(value, shots=4096)
```

- **Note:** `even_parity_ground_state` uses exact diagonalization – not VQE. The pseudocode masks this gap; a real VQE loop is the next milestone.
- Exact diagonalization is valid only for small  $L$ ; it serves as a validation benchmark, not a scalable method.
- The measured quantity is the same Pauli string a circuit would estimate from bitstrings.
- The same workflow can be executed on quantum hardware using repeated circuit measurements.
- The next code revision should plug this sweep into the VQE runner directly.

# Main Conclusions

- The edge Pauli string is a measurable finite-size diagnostic of Kitaev topology.
- Exact diagonalization shows string enhancement alongside near-degenerate parity sectors.
- Synthetic shot sampling reproduces the ideal string-correlation trend.
- A real VQE sweep and hardware-noise study remain the next validation steps.

## Conceptual Message

Local observables may change across the transition, but parity-sector degeneracy and non-local edge coherence identify the topological phase.

## Next: Block 4

- Add a real VQE loop for every  $\mu$  point instead of ED-prepared validation states.
- Attach depolarizing, readout, and relaxation noise models to the measurement circuits.
- Compare ideal, noisy, and mitigated string-order curves.
- Study robustness of non-local topology diagnostics under realistic imperfections.
- Investigate how much noise the phase boundary tolerates before the signal becomes unreliable.
- Extend sweep to a 2D  $(t, \mu)$  phase diagram.